

8. TROUBLESHOOTING

8-1. Troubleshooting Summary

1. Before troubleshooting

1) Applied models

① SMMS (i) Multi type models

Indoor unit : MMX-APXXX,

Outdoor unit : MMY-MAPXXXXT8X, MMY-MAPXXXHT7X

② Super Heat Recovery Multi type models

Indoor unit : MMX-APXXX,

Outdoor unit : MMY-MAPXXXFT8X

③ Mini-SMMS Multi type models

Indoor unit : MMX-APXXX,

Outdoor unit : MCY-MAPXXXHT, MCY-MAPXXXHT2X

2) Required tools / measuring devices

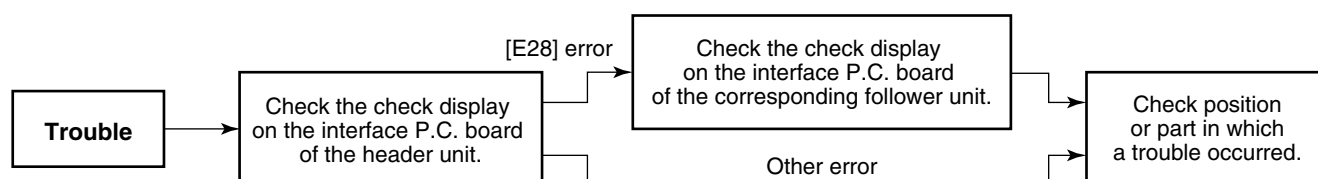
- Screwdrivers (Philips, Minus), spanner, radio pinchers, nipper, push pin for reset switch, etc.
- Tester, thermometer, pressure gauge, etc.

3) Confirmation before check (The following items are not troubles.)

No.	Operation	Check items
1	Compressor does not operate.	• Is not delayed for 3 minutes? (3 minutes after compressor-OFF) • Is not thermostat OFF? • Is not the fan operating or timer? • Is not the system initially communicating? Heating operation cannot be performed under condition of outside temperature 21°C or higher. Cooling operation cannot be performed under condition of outside temperature -5°C or lower.
2	Indoor fan does not work.	• Is not the cold draft prevention being controlled in heating operation?
3	Outdoor fan does not rotate, or fan speed changes.	• Is not low cooling operation being controlled? • Is not a defrost operation being performed?
4	Indoor fan does not stop.	• Is not after-heat elimination operation being controlled after heating operation?
5	Start/stop operation on remote controller is unavailable.	• Is not auxiliary unit or remote control being operated?
6	—	• Is connecting wire of indoor unit or remote controller correct?

2. Troubleshooting procedure

When a trouble occurred, advance the check operation in the following procedure.



NOTE

While a check operation is performed, a malfunction of the microprocessor may be caused due to condition of the power supply or the external noise.

If there is any noise source, change wires of the remote controller and signal wires to shield wires.

8-2. How to check

On the remote controller (Wired remote controller, Central control remote controller) and on the interface P.C. board of the outdoor unit, LCD display part (Remote controller) or 7-segment display part (on outdoor interface P.C. board) is provided in order to display the operation status.

When a trouble occurred, the method to judge the trouble or defective position of the air conditioner by this self-diagnosis function is shown below.

The following table shows the list of each check code that each device detects. Check the check contents in the following table according to position to be checked.

- Check from the indoor remote controller or TCC-LINK central controller: Refer to “Display on remote controller & TCC-LINK central controller” in the following table.
- Check from outdoor unit: Refer to “Display of outdoor segment” in the following table.
- Check from indoor unit of wireless remote controller: Refer to Receiver lamp display” in the following table.

Check code display list (Indoor unit)

(*) ○: Goes on, ◎: Flashes, ●: Goes off

A (Alternate) : Flashing condition is alternate when there are two flashing LED.

S (Simultaneously) : Two LED flash simultaneously when there are two flashing LED.

[Indoor unit detects error.]

Check code display			Receiver lamp display				Main defective position	Description
TCC-LINK central & remote controller	Outdoor 7-segment		Block display (*)					
		Auxiliary code	Operation	Timer	Ready	Flash		
E03	—	—	◎	●	●		Regular communication error between indoor and remote controller	No communication from remote controller and network adapter (No central control system communication also)
E04	—	—	●	●	◎		Regular communication error between indoor and outdoor	No communication from outdoor unit
E08	E08	Duplicated indoor unit No.	◎	●	●		Duplicated indoor address	An address same to self address was detected.
E11	—	—	◎	●	●		Communication error between Application control kit and indoor unit	Communication error between Application control kit and indoor unit P.C. board
E18	—	—	◎	●	●		Regular communication error between header and follower in indoor unit	Regular communication between header and follower units in indoor unit was impossible.
F01	—	—	◎	◎	●	A	Indoor heat exchanger temp. sensor (TCJ) error	Open/short of heat exchanger temp. sensor (TCJ) was detected.
F02	—	—	◎	◎	●	A	Indoor heat exchanger temp. sensor (TC2) error	Open/short of heat exchanger temp. sensor (TC2) was detected.
F03	—	—	◎	◎	●	A	Indoor heat exchanger temp. sensor (TC1) error	Open/short of heat exchanger temp. sensor (TC1) was detected.
F10	—	—	◎	◎	●	A	Room tem. Sensor (TA) error	Open/short of room temp. sensor (TA) was detected.
F29	—	—	◎	◎	●	S	Indoor or other P.C. board error	Indoor EEPROM error (Other error may be detected.)
L03	—	—	◎	●	◎	S	Duplicated setting of header in indoor group	There were multiple header units in a group.
L07	—	—	◎	●	◎	S	There is group cable in individual indoor unit.	There is even an indoor unit connected to group in individual indoor unit.
L08	L08	—	◎	●	◎	S	Indoor group address is unset.	Indoor group address is unset. (Detected also at outdoor unit side)
L09	—	—	◎	●	◎	S	Indoor capacity is unset.	Capacity of indoor unit is unset.
L20	—	—	◎	○	◎	S	Duplicated central control system address	Setting of central control system address is duplicated.
L30	L30	Detected indoor unit No.	◎	○	◎	S	External error was input in indoor (Interlock).	System abnormally stopped by input of external error (CN80).
P01	—	—	●	◎	◎	A	Indoor AC fan error	Error of indoor AC fan was detected. (Fan motor thermal relay operation)
P10	P10	Detected indoor unit No.	●	◎	◎	A	Indoor overflow was detected.	Float switch operated.
P12	—	—	●	◎	◎	A	Indoor DC fan error	Error (Over-current, lock, etc.) of indoor DC fan was detected.
P31	—	—	◎	●	◎	A	Other indoor unit error	Group follower unit cannot be operated by [E03/L03/L07/L08] alarm of header unit.

Note) The check code display may be different according to the detected device even same error contents such as communication error.

[Remote controller detects error.]

(*) ○: Goes on, ◎: Flashes, ●: Goes off

A (Alternate) : Flashing condition is alternate when there are two flashing LED.

S (Simultaneously) : Two LED flash simultaneously when there are two flashing LED.

Check code display			Receiver lamp display				Main defective position	Description
Remote controller	Outdoor 7-segment		Block display (*)					
		Auxiliary code	Operation	Timer	Ready	Flash		
E01	—	—	◎	●	●		No remote controller header unit, remote controller communication (receive) error	When signal cannot be received from indoor unit, when header of remote controller was not set (including 2 remote controllers)
E02	—	—	◎	●	●		Remote controller communication (send) error	When signal cannot be sent to indoor unit
E09	—	—	◎	●	●		Duplicated remote controller header	In 2-remote controller control, both remote controllers were set to header. (Indoor header stops with alarm and follower unit continues operation.)

[Central controller detects error.]

Check code display			Receiver lamp display				Main defective position	Description
TCC-LINK central	Outdoor 7-segment		Block display (*)					
		Auxiliary code	Operation	Timer	Ready	Flash		
C05	—	—	Is not displayed (In shared use of remote controller)				Central control system communication (send) error	When signal of central control system cannot be sent, there are same multiple central devices (AI-NET)
C06	—	—					Central control system communication (receive) error	When signal of central control system cannot be received
—	—	—					There are multiple network adapters.	There were multiple network adapters (AI-NET) on remote controller communication line.
C12	—	—	—				Batched alarm of interface for general-purpose equipment control	Error of equipment connected to control interface of the general-purpose unit exclusive to TCC-LINK/AI-NET
P30	—	—	According to unit with alarm (Abovementioned)				Group follower unit error	Group follower unit error (For remote controller, [***] details is displayed together with unit No.)

Note) The check code display may be different according to the detected device even same error contents such as communication error.

List of check codes (outdoor unit)

(Errors detected by SMMS outdoor interface - typical examples)

IPDU: Intelligent Power Drive Unit (Inverter P.C. board)

○ : Lighting, ◎ : Flashing, ● : Goes off

ALT.: Flashing is alternately when there are two flashing LED

SIM: Simultaneous flashing when there are two flashing LED

Check code			Display of receiving unit				Typical fault site	Description of error																																																																																																															
Outdoor 7-segment display			TCC-LINK central control or main remote controller display	Indicator light block																																																																																																																			
	Sub-code			Operation ⏻	Timer ⌚	Ready ⦿			Flash																																																																																																														
E06	Number of indoor units from which signal is received normally		E06	●	●	⦿		Dropping out of indoor unit	Indoor unit initially communicating normally fails to return signal (reduction in number of indoor units connected).																																																																																																														
E07	—		(E04)	●	●	⦿		Indoor-outdoor communication circuit error	Signal cannot be transmitted to indoor units (→ indoor units left without communication from outdoor unit).																																																																																																														
E08	Duplicated indoor address		(E08)	⦿	●	●		Duplicated indoor address	More than one indoor unit is assigned same address (also detected at indoor unit end).																																																																																																														
E12	01: Indoor-outdoor communication 02: Outdoor-outdoor communication		E12	⦿	●	●		Automatic address starting error	- Indoor automatic address setting is started while automatic address setting for equipment in other refrigerant line is in progress. - Outdoor automatic address setting is started while automatic address setting for indoor units is in progress.																																																																																																														
E15	—		E15	●	●	⦿		Indoor unit not found during automatic address setting	Indoor unit fails to communicate while automatic address setting for indoor units is in progress.																																																																																																														
E16	00: Overloading 01: Number of units connected		E16	●	●	⦿		Too many indoor units connected / overloading	Combined capacity of indoor units is too large (more than 135 % of combined capacity of outdoor units).																																																																																																														
E19	00: No header unit 02: Two or more header units		E19	●	●	⦿		Error in number of outdoor header units	There is no or more than one outdoor header unit in one refrigerant line.																																																																																																														
E20	01: Connection of outdoor unit from other refrigerant line 02: Connection of indoor unit from other refrigerant line		E20	●	●	⦿		Connection to other refrigerant line found during automatic address setting	Indoor unit from other refrigerant line is detected while indoor automatic address setting is in progress.																																																																																																														
E23	—		E23	●	●	⦿		Outdoor-outdoor communication transmission error	Signal cannot be transmitted to other outdoor units.																																																																																																														
E25	—		E25	●	●	⦿		Duplicated follower outdoor address	There is duplication in outdoor addresses set manually.																																																																																																														
E26	Address of outdoor unit from which signal is not received normally		E26	●	●	⦿		Dropping out of outdoor unit	Follower outdoor unit initially communicating normally fails to do so (reduction in number of follower outdoor units connected).																																																																																																														
E28	Detected outdoor unit No.		E28	●	●	⦿		Outdoor follower unit error	Outdoor header unit detects fault relating to follower outdoor unit (detail displayed on follower outdoor unit).																																																																																																														
E31	<table><tr><td></td><td colspan="3">A3-IPDU</td><td>Fan</td><td></td><td colspan="3">A3-IPDU</td><td>Fan</td></tr><tr><td></td><td>1</td><td>2</td><td>3</td><td>IPDU</td><td></td><td>1</td><td>2</td><td>3</td><td>IPDU</td></tr><tr><td>01</td><td>○</td><td></td><td></td><td></td><td>0A</td><td>○</td><td></td><td></td><td>○</td></tr><tr><td>02</td><td></td><td>○</td><td></td><td></td><td>0B</td><td>○</td><td>○</td><td></td><td>○</td></tr><tr><td>03</td><td>○</td><td>○</td><td></td><td></td><td>0C</td><td></td><td></td><td>○</td><td>○</td></tr><tr><td>04</td><td></td><td></td><td>○</td><td></td><td>0D</td><td>○</td><td></td><td>○</td><td>○</td></tr><tr><td>05</td><td>○</td><td></td><td>○</td><td></td><td>0E</td><td></td><td>○</td><td>○</td><td>○</td></tr><tr><td>06</td><td></td><td>○</td><td>○</td><td></td><td>0F</td><td>○</td><td>○</td><td>○</td><td>○</td></tr><tr><td>07</td><td>○</td><td>○</td><td>○</td><td></td><td colspan="5">Circle (○): Faulty IPDU</td></tr><tr><td>08</td><td></td><td></td><td></td><td>○</td><td colspan="5"></td></tr><tr><td>09</td><td>○</td><td></td><td></td><td>○</td><td colspan="5"></td></tr></table>			A3-IPDU			Fan		A3-IPDU			Fan		1	2	3	IPDU		1	2	3	IPDU	01	○				0A	○			○	02		○			0B	○	○		○	03	○	○			0C			○	○	04			○		0D	○		○	○	05	○		○		0E		○	○	○	06		○	○		0F	○	○	○	○	07	○	○	○		Circle (○): Faulty IPDU					08				○						09	○			○						E31	●	●	⦿		IPDU communication error	There is no communication between IPDUs (P.C. boards) in inverter box.
	A3-IPDU			Fan		A3-IPDU			Fan																																																																																																														
	1	2	3	IPDU		1	2	3	IPDU																																																																																																														
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F04	—		F04	⦿	⦿	○	ALT	Outdoor discharge temperature sensor (TD1) error	Outdoor discharge temperature sensor (TD1) has been open / short-circuited.																																																																																																														
F05	—		F05	⦿	⦿	○	ALT	Outdoor discharge temperature sensor (TD2) error	Outdoor discharge temperature sensor (TD2) has been open / short-circuited.																																																																																																														
F06	01: TE1 02: TE2		F06	⦿	⦿	○	ALT	Outdoor heat exchanger temperature sensor (TE1, TE2) error	Outdoor heat exchanger temperature sensors (TE1, TE2) have been open / short-circuited.																																																																																																														
F07	—		F07	⦿	⦿	○	ALT	Outdoor liquid temperature sensor (TL) error	Outdoor liquid temperature sensor (TL) has been open / short-circuited.																																																																																																														
F08	—		F08	⦿	⦿	○	ALT	Outdoor outside air temperature sensor (TO) error	Outdoor outside air temperature sensor (TO) has been open / short-circuited.																																																																																																														
F11	—		F11																																																																																																																				

Check code			Display of receiving unit				Typical fault site	Description of error
Outdoor 7-segment display		TCC-LINK central control or main remote controller display	Indicator light block					
	Sub-code		Operation 	Timer 	Ready 	Flash 		
F12	—	F12				ALT	Outdoor suction temperature sensor (TS1) error	Outdoor suction temperature sensor (TS1) has been open / short-circuited.
F13	01: Compressor 1 02: Compressor 2 03: Compressor 3	F13				ALT	Outdoor IGBT built-in temperature sensor (TH) error	Open-circuit or short-circuit of the outdoor IGBT built-in temperature sensor (TH) was detected.
F15	—	F15				ALT	Outdoor temperature sensor (TE1, TL) wiring error	Wiring error in outdoor temperature sensors (TE1, TL) has been detected.
F16	—	F16				ALT	Outdoor pressure sensor (Pd, Ps) wiring error	Wiring error in outdoor pressure sensors (Pd, Ps) has been detected.
F22	—	F22				ALT	Outdoor discharge temperature sensor (TD3) error	Outdoor discharge temperature sensor (TD3) has been open / short-circuited.
F23	—	F23				ALT	Low pressure sensor (Ps) error	Output voltage of low pressure sensor (Ps) is zero.
F24	—	F24				ALT	High pressure sensor (Pd) error	Output voltage of high pressure sensor (Pd) is zero or provides abnormal readings when compressors have been turned off.
F31	—	F31				SIM	Outdoor EEPROM error	Outdoor EEPROM is faulty (alarm and shutdown for header unit and continued operation for follower unit)
H01	01: Compressor 1 02: Compressor 2 03: Compressor 3	H01					Compressor breakdown	Overcurrent of the inverter current (Idc) detection circuit was detected.
H02	01: Compressor 1 02: Compressor 2 03: Compressor 3	H02					Compressor error (Lock)	Compressor lock was detected.
H03	01: Compressor 1 02: Compressor 2 03: Compressor 3	H03					Current detection circuit error	Current error was detected while the compressor was stopped.
H04		H05					Compressor 1 case thermo activation	Compressor 1 case thermo was activated for protection.
H05	—	H05					Outdoor discharge temperature sensor (TD1) wiring error	Wiring / installation error or detachment of outdoor discharge temperature sensor (TD1) has been detected.
H06	—	H06					Activation of low-pressure protection	Low pressure (Ps) sensor detects abnormally low operating pressure.
H07	—	H07					Activation of low-pressure protection	Temperature sensor for oil level detection (TK1-5) detects abnormally low oil level.
H08	01: TK1 sensor error 02: TK2 sensor error 03: TK3 sensor error 04: TK4 sensor error 05: TK5 sensor error	H08					Error in temperature sensor for oil level detection (TK1-5)	Temperature sensor for oil level detection (TK1-5) has been open / short-circuited.
H14	—	H14					Compressor 2 case thermo activation	Compressor 2 case thermo was activated for protection.
H15	—	H15					Outdoor discharge temperature sensor (TD2) wiring error	Wiring / installation error or detachment of outdoor discharge temperature sensor (TD2) has been detected.
H16	01: TK1 oil circuit error 02: TK2 oil circuit error 03: TK3 oil circuit error 04: TK4 oil circuit error 05: TK5 oil circuit error	H16					Oil level detection circuit error	No temperature change is detected by temperature sensor for oil level detection (TK1-5) despite compressor having been started.
H25	—	H25					Outdoor discharge temperature sensor (TD3) wiring error	Wiring / installation error or detachment of outdoor discharge temperature sensor (TD3) has been detected.
L04	—	L04				SIM	Duplicated outdoor refrigerant line address	Identical refrigerant line address has been assigned to outdoor units belonging to different refrigerant piping systems.
L06	Number of priority indoor units (check code L05 or L06 depending on individual unit)	L05				SIM	Duplicated priority indoor unit (as displayed on priority indoor unit)	More than one indoor unit has been set up as priority indoor unit.
		L06				SIM	Duplicated priority indoor unit (as displayed on indoor unit other than priority indoor unit)	More than one indoor unit has been set up as priority indoor unit.

Check code				Display of receiving unit				Typical fault site	Description of error	
Outdoor 7-segment display				TCC-LINK central control or main remote controller display	Indicator light block					
	Sub-code				Operation ⏻	Timer ⌚	Ready ⚙️			Flash
L08	—			(L08)	⊙	●	⊙	SIM	SIM Indoor group address not set	Address setting has not been performed for one or more indoor units (also detected at indoor end).
L10	—			L10	⊙	○	⊙	SIM	Outdoor capacity not set	Outdoor unit capacity has not been set (after P.C. board replacement).
L17	—			L17	⊙	○	⊙	SIM	Outdoor model incompatibility error	Old model outdoor unit (prior to 3 series) has been connected.
L18	—			L18	⊙	○	⊙	SIM	FS (Flow Selector) unit error	Cooling / heating cycle error resulting from piping error is detected.
L28				L28	⊙	○	⊙	SIM	Too many outdoor units connected	More than four outdoor units have been connected.
L29	SMMS (Series 1) 01: A3-IPDU1 error 02: A3-IPDU2 error 03: A3-IPDU1/A3-IPDU2 error 04: Fan IPDU error 05: A3-IPDU1 + Fan IPDU error 06: A3-IPDU2 + Fan IPDU error 07: All IPDU error SMMS-i (Series 4)			L29	⊙	○	⊙	SIM	Error in number of IPDUs	There are insufficient number of IPDUs (P.C. boards) in inverter box.
		A3-IPDU 1 2 3	Fan IPDU		A3-IPDU 1 2 3	Fan IPDU				
	01	○			0A	○		○		
	02	○			0B	○	○		○	
	03	○	○		0C			○	○	
	04		○		0D	○		○	○	
	05	○		○	0E	○	○	○	○	
	06		○	○	0F	○	○	○	○	
	07	○	○	○				Circle (○): Faulty IPDU		
	08				○					
	09	○			○					
L30	Detected indoor unit No.			(L30)	⊙	○	⊙	SIM	Indoor external error input (interlock)	Indoor unit has been shut down for external error input in one refrigerant line (detected by indoor unit).
P03	—				⊙	●	⊙	ALT	Outdoor discharge (TD1) temperature error	Outdoor discharge temperature sensor (TD1) has detected abnormally high temperature.
P04	P0301: Compressor 1 02: Compressor 2 03: Compressor 3			P04	⊙	●	⊙	ALT	High-pressure SW activation	High-pressure SW was activated.
P05	00: Open phase detected			P05	⊙	●	⊙	ALT	Open phase / power failure	Open phase is detected when power is turned on. Inverter DC voltage is too high (overvoltage) or too low (undervoltage).
	01: Compressor 1 02: Compressor 2 03: Compressor 3								Inverter DC voltage (Vdc) error MG-CTT error	
P07	01: Compressor 1 02: Compressor 2 03: Compressor 3			P07	⊙	●	⊙	ALT	Heat sink overheating error	Temperature sensor built into IGBT (TH) detects overheating.
P10	Indoor unit No. detected			(P10)	●	⊙	⊙	ALT	Indoor unit overflow	Indoor unit has been shutdown in one refrigerant line due to detection of overflow (detected by indoor unit).
P13	—			P13	●	⊙	⊙	ALT	Outdoor liquid backflow detection error	State of refrigerant cycle circuit indicates liquid backflow operation.
P15	01: TS condition 02: TD condition			P15	⊙	●	⊙	ALT	Gas leak detection	Outdoor suction temperature sensor (TS1) detects sustained and repeated high temperatures that exceed standard value.
P17	—			P17	⊙	●	⊙	ALT	Outdoor discharge (TD2) temperature error	Outdoor discharge temperature sensor (TD2) detects abnormally high temperature.
P18	—			P18	⊙	●	⊙	ALT	Outdoor discharge (TD3) temperature error	Outdoor discharge temperature sensor (TD3) detects abnormally high temperature.
P19	Outdoor unit No. detected			P19	⊙	●	⊙	ALT	4-way valve reversing error	Abnormality in refrigerating cycle is detected during heating operation.
P20	—			P20	⊙	●	⊙	ALT	Activation of high-pressure protection	High pressure (Pd) sensor detects high pressure that exceeds standard value.

MG-CTT: Magnet contactor